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Chapter 1

Introduction

1.1 Motivation

Financial time series data is never fully observed. Limit-order book feeds drop quotes during bursts of order cancellations. Equity markets suspend trading when prices move too violently. Over-the-counter bond markets leave entire yield-curve tenors unquoted for days. Implied volatility surfaces contain large regions where no liquid option trade has occurred. Macro and alternative data arrive at mixed, irregular frequencies, with publication lags that create structural gaps at the daily level. This missing data is pervasive and often elusive, and gets in the way of advanced analysis. Thus practitioners or a researchers face the same decision: what value should be attributed to the missing observation, and how much uncertainty should be attached to that attribution?

The simplest and most common solution is to carry forward the last observed value, but this can lead to systematic mispricing, underestimation of risk, and distortion of correlations among assets.

We aim to take a different path, one that looks for solutions that aim to preserve as much as possible the statistical properties of the series, in order to maintain the integrity of the underlying data and consequently the robustness of any downstream model that relies on it. Since this can be considered an inference problem, we will also explore ML algorithms, and in classical machine learning approach, we will focus on quality of inference rather than interpretability of the model.

In this thesis we will test these different methods of imputation, and evaluate them not only on the accuracy of the imputation itself (MSE,...), but also on the quality of the downstream model that uses the imputed data, and how far it deviates from the case where all data is observed.

1.2 Literature review

The literature on missing data is vast and spread across statistics, econometrics and, more recently, machine learning. Rather than attempting an exhaustive survey, we trace the ideas this thesis directly builds on: the foundational framework that defines what missingness is, the classical likelihood-based methods with the EM algorithm at their core, the progressive shift towards algorithmic and deep learning imputers, and finally how these tools have been absorbed by economics and finance.

1.2.1 Foundations

Until the mid-1970s missing data was mostly treated as a nuisance to be engineered away: incomplete records were dropped, or filled in with some convenient value, and the analysis proceeded as if nothing had happened. The turning point is Rubin (1976), who proposed to treat missingness itself as a random object. Given a data matrix and an indicator matrix flagging which entries are observed, the joint distribution factorizes into the law of the data and the law of the *missingness mechanism*, the conditional distribution of the indicators given the data. Rubin's classification of this mechanism is still the standard language of the field: data are missing completely at random (MCAR) when the mechanism is independent of the data; missing at random (MAR) when it depends only on the observed part; and missing not at random (MNAR) otherwise. The central result of the paper is *ignorability*: under MAR, plus a parameter-distinctness condition, likelihood-based and Bayesian inference can ignore the mechanism altogether and work with the observed-data likelihood. This single result justifies most of the methods in this thesis; whenever it fails, as in the stressed missingness pattern of Section 2.1.1, every method that relies on it stands on shaky ground.

The monograph of Little and Rubin (2019), first published in 1987, organized the field around this framework and remains its canonical reference. Their taxonomy of approaches (complete-case analysis, weighting, single imputation, likelihood-based methods) is still the way the subject is taught, and their analysis of the simple methods is directly relevant here: listwise deletion is unbiased only under MCAR and can discard a dramatic fraction of the sample, while unconditional mean imputation distorts variances and correlations even under MCAR. These are precisely the two baselines we adopt in Chapter 2, and the literature predicts their failure modes quite exactly.

Two other foundational strands deserve mention. The first is weighting: instead of imputing, one reweights the complete cases by the inverse of their probability of being observed. Robins et al. (1994) developed this idea into a full semiparametric theory, including doubly robust estimators; Seaman and White (2013) give an accessible review. The second is *multiple imputation*, introduced by Rubin (1987): draw several imputed datasets from the posterior predictive distribution of the missing values, analyse each one as if it were complete, and combine the results with simple rules that propagate the imputation uncertainty into the final standard errors (see also Rubin, 1996). Multiple imputation is the historical answer to a criticism that applies to every single-imputation method in this thesis: an imputed dataset treated as real understates uncertainty, because imputed values are estimates, not observations.

The MNAR case is fundamentally harder, since the mechanism is not identifiable from the observed data alone and any analysis rests on untestable assumptions. In econometrics the problem was attacked early and largely independently by Heckman (1979), whose sample-selection model corrects for non-randomly missing outcomes through an explicit model of the selection process; it remains the standard MNAR tool in applied economics. In biostatistics the accepted practice is sensitivity analysis across plausible mechanisms (Molenberghs and Kenward, 2007). Finally, the translation of MAR to dependent, dynamically evolving data is more subtle than it appears; Commenges and Jacqmin-Gadda (2022) give a rigorous stochastic-process formulation, which is the natural setting for financial time series.

1.2.2 Likelihood methods and the EM algorithm

Once ignorability is granted, inference reduces to maximizing the observed-data likelihood, which almost never has a closed form because the missing entries must be integrated out. The Expectation–Maximization algorithm of Dempster et al. (1977) is the fundamental tool for this task, and arguably the most important paper in the field after Rubin (1976). Its logic mirrors the structure of the problem: the E-step computes the conditional expectation of the complete-data log-likelihood given the observed data and the current parameters, and the M-step maximizes it. The paper proved that each iteration increases the observed likelihood, unified a large number of pre-existing special-purpose schemes, and named the method. The convergence theory was completed by Wu (1983), who fixed a flaw in the original proof and gave the regularity conditions under which EM converges to a stationary point. Since EM never forms the observed information matrix, standard errors require extra work; Louis (1982) showed how to recover them from complete-data quantities. Among the many refinements that followed, the ECM and ECME algorithms (Liu and Rubin, 1994; Meng and Rubin, 1993) replace the M-step with cheaper conditional maximizations; McLachlan and Krishnan (2008) is the standard book-length treatment.

For our purposes the relevant instance is EM for the mean vector and covariance matrix of an incomplete multivariate normal sample (Little and Rubin, 2019, Ch. 11): the E-step reduces to linear regressions of the missing coordinates on the observed ones, and the fixed point delivers both the maximum likelihood estimate of $(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ and, as a by-product, conditional-expectation imputations. Since daily returns are decidedly non-Gaussian (Cont, 2001), the robust variant matters: Little (1988) derived EM for the multivariate Student-t, whose E-step simply downweights outlying observations, and Liu and Rubin (1995) made its estimation practical via ECME. This is the version we implement in Chapter 2.

The time series counterpart of this machinery is the state-space framework. The Kalman filter (Kalman, 1960) accommodates missing observations with no modification at all: when a measurement is absent, the update step is skipped and the prediction carries through. Jones (1980) exploited this to fit ARMA models to incomplete series by exact maximum likelihood, and Shumway and Stoffer (1982) closed the loop with EM: the Kalman smoother computes exactly the conditional expectations the E-step requires, so parameter estimation and imputation come out of the same iteration. Harvey and Pierse (1984) applied these ideas to economic time series, Kohn and Ansley (1986) settled exact interpolation for ARIMA models, and the monographs of Harvey (1989) and Durbin and Koopman (2012) remain the references for the general theory.

1.2.3 From multiple imputation to machine learning

On the applied side, the 1990s and 2000s standardized imputation practice. Schafer (1997) provided workable algorithms for multiple imputation under joint multivariate models, and the chained equations approach of van Buuren and Groothuis-Oudshoorn (2011) replaced the joint model with a sequence of univariate conditional models, one per variable, cycled until convergence; its flexibility made it the de facto standard in biostatistics and the social sciences. van Buuren (2018) gives a modern account of the whole practice.

In parallel, a more algorithmic tradition developed methods that dispense with an explicit probability model. k-nearest-neighbours imputation was popularized by Troyanskaya et al. (2001) for gene expression data and transfers directly to return panels: a missing entry is filled with a weighted average of the corresponding entries of similar rows (dates, in our case); random forests followed (Stekhoven and Bühlmann, 2012). A distinct and elegant line is low-rank matrix completion (Candès and Recht, 2009; Mazumder et al., 2010), which recovers missing entries under the assumption that the data matrix is approximately low-rank, a natural assumption for equity returns where a few factors explain most of the covariance.

Deep learning entered the field once recurrent networks were adapted to irregular inputs. GRU-D (Zhengping Che and Liu, 2016) feeds the missingness mask and the time since the last observation into a gated recurrent unit with learned decay; BRITS (Cao et al., 2018) treats imputations as variables of a bidirectional recurrent graph, trained by consistency between the two directions, and requires no assumption on the generating process. Generative approaches recast imputation as conditional sampling: GAIN adapts adversarial training (Yoon et al., 2018), GP-VAE combines a variational autoencoder with a Gaussian process prior in latent space (Fortuin et al., 2020), SAITS relies on self-attention (Du et al., 2023), and CSDI (Tashiro et al., 2021) uses conditional score-based diffusion, producing a full predictive distribution for the missing values rather than a point estimate, which is why we select it as our generative representative. Dong et al. (2024) survey this rapidly growing area, and Kreiss et al. (2024) bridge it with the signal processing literature.

Two caveats recur across this literature. First, essentially all deep imputers assume MAR, usually implicitly through their masked training objective; identification under more general mechanisms has been studied only very recently (Ma et al., 2025). Second, evaluation is almost universally pointwise, RMSE or MAE on artificially masked entries, which says little about whether the joint distribution, and in particular the covariance structure, survives imputation. Both caveats are starting points for this thesis.

1.2.4 Applications in economics and finance

In economics, missing data rarely goes by that name. The oldest example is non-synchronous trading: Scholes and Williams (1977) showed that stale prices, which are in effect missing prices carried forward, bias beta estimates, and derived corrected estimators. This is exactly the last-observation-carried-forward distortion described in Section 1.1.

The macroeconometric literature turned missing data handling into routine practice, because macro datasets are incomplete by construction: series are released at mixed frequencies and with staggered publication lags (the so-called ragged edge). Stock and Watson (2002) used an EM–principal-components iteration to extract factors from large unbalanced panels; Mariano and Murasawa (2003) treated quarterly GDP as a monthly series with two out of three values missing, estimated in state-space form; Giannone et al. (2008) formalized nowcasting, where the Kalman filter absorbs data releases as they arrive; and Doz et al. (2012) and Bańbura and Modugno (2014) developed maximum likelihood estimation of large factor models under arbitrary missingness patterns, essentially scaling Shumway and Stoffer (1982) to hundreds of series. This machinery runs in production at central banks today. For panels with a time dimension, Honaker and King (2010) adapted multiple imputation to time-series cross-section data, noting that off-the-shelf procedures ignore temporal smoothness.

In finance proper, the seminal reference is Stambaugh (1997), who considered assets whose return histories start at different dates, the typical shape of any real multi-asset panel, and derived maximum likelihood estimators of means and covariances that use the long histories to sharpen the moments of the short ones. The message, that discarding incomplete history throws away information about the covariance matrix, is a direct ancestor of our evaluation criteria. Kofman and Sharpe (2003) later imported multiple imputation into empirical finance, showing on several canonical applications that complete-case analysis materially biases the conclusions.

The topic has recently returned to the centre of asset pricing, driven by the firm-characteristics panels used in cross-sectional factor models, where the majority of entries can be missing. Bryzgalova et al. (2025) document that the profession's ad hoc fixes, typically cross-sectional medians, are not innocuous: the choice of imputation changes the estimated risk premia, and they propose a factor-based imputation exploiting both cross-sectional and time series dependence. In parallel, Freyberger et al. (2025) show that firm characteristics are missing *not* at random and develop estimators that remain consistent under such mechanisms. Both papers make, in the cross-section of characteristics, the point this thesis makes for return panels: what matters is not the imputation error itself but its effect on the downstream object of interest. Finally, Choudhury et al. (2021) highlight a specifically financial pitfall: imputing with a model trained on the full sample leaks future information into the imputed values, and they formalize the resulting trade-off between look-ahead bias and imputation variance.

Summing up, the statistical literature provides the taxonomy of mechanisms and the classical estimators, the machine learning literature provides increasingly powerful conditional models trained under an implicit MAR assumption, and the finance literature provides both the warning that missingness is often MNAR precisely when it matters most (crises, illiquidity) and the reminder that imputation should be judged by the downstream model. The experimental design of this thesis, controlled injection of MCAR and MNAR patterns into a complete equity panel, classical and deep imputers side by side, and evaluation on covariance geometry, risk measures and portfolio construction rather than pointwise error alone, sits deliberately at the intersection of these three strands.

1.3 Research questions

1.4 Contributions

1.5 Outline of the thesis

Chapter 2

Methodology

As already mentioned, missing data can arise in countless different ways, and the choice of methodology depends on the nature of the missingness, the type of data, and the specific goals of the analysis. We will focus on a subset of cases, namely those that involve imputation of the missing values with "information" from the whole dataset (no causal constraint), with in mind the goal of training a downstream model that is as close as possible to the one that would have been trained on the complete dataset. Our choice of data, models, and evaluation methods will be guided by these constraints. We will nevertheless mention when/how models should be tweaked to solve adjacent problems.

2.1 Data

As our main baseline, we chose the most studied and reliable dataset in finance: the end-of-day prices of the S&P500, in this case fetched with yfinance from 1/1/2010 to 30/5/2026 (potrei cambiare le date). It clearly is not a dataset inflicted with the curse of missing data; the only structural ones being occasional circuit breaks and listings/delistings from the index. To obtain a full dataset with no "real" missing data, we removed each stock that contained at least one missing observation. Furthermore, we selected a sample of 50 stocks from the index, stratified across industries; this is a sensible choice because:

1. with more 400 stocks the covariance matrix $\mathbf{\Sigma}$ quickly becomes unreliable (noisy covariances, skewed eigenvalues, unstable inversion. . .) as the number of observations shrinks;

2. it is easier to check whether the ground data is actually correct;

3. model training requires considerably less memory, in a time where RAM and GPU prices are skyrocketing;

4. it is well known that a well constructed portfolio with more than 30 stocks is already sufficiently diversified (since volatility scales with the inverse of the square root of

the number of securities).

This dataset will serve as the ground truth against which we will compare our results. In parallel, we will modify it by inserting fake missing observations according to different rules, as we define in the subsequent sections.

Now an important choice is whether to operate in the prices "space" P_t or the log returns space R_t. There are positives and negatives about each. For one, the raw dataset is prices, and in practice missing data will mostly come in the form of missing prices. Furthermore, in most applications we care slightly more about prices than returns, since the formers are what is actually bought and sold. But P_t has the drawback of not being a complete space ($p_{t,k} > 0$) and being highly level dependant, making the series non-stationary and very difficult for ML algorithms to handle. We will thus focus on R_t.

2.1.1 Missingness Injection

(Da scrivere meglio, questa è solo l'outline)

As outlined in the literature review ...

We used three algorithms to simulate different missingness patterns:

1. **Sparse**: each scalar $r_{t,k}$ is missing with probability r, independent of the rest of the dataset. This is the easiest example of MCAR and can simulate a randomly corrupted dataset;

2. **Contiguous gaps**: n points $r_{t,k}$ are chosen at random and eliminated. Then subsequent observations of the same variable $r_{t+1,k}, \dots, r_{t+h,k}$ get deleted according to an exponential decay with rate λ. This is still an example of MCAR and can be thought to simulate assets failing to trade because of random shortages of liquidity;

3. **Stressed**: we calculate index squared returns for each date t and rank them. Then, returns whose date is above a certain percentile $p$get deleted with probability h, while the rest still go missing with rate r. This simulates shortages of liquidity or circuit breaks that happen when markets experience high levels of volatility; since the missingness of a given $r_{t,k}$ is injected conditional on all the other values of the dataset, this is an example of MNAR.

For each of the three missingness patterns, we will vary the parameters r, λ, and h to simulate different levels of missingness. Furthermore, we will make it so that the ratio of missing values is the same for each level of missingness in each pattern, so that we can compare the performance of the different imputation methods in a sensible way. The exact values of the parameters are reported in Table 2.2. The column "Rate" reports the expected value of the missingness ratio for each pattern.

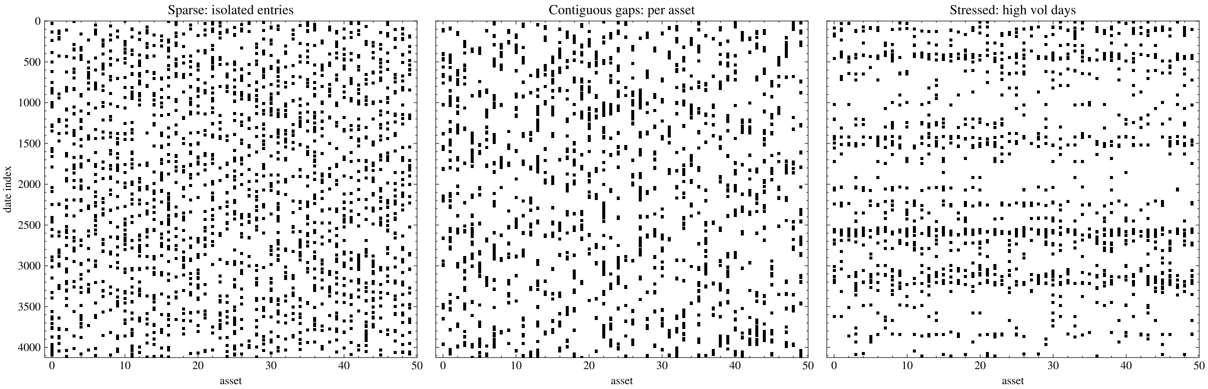

Figure 2.1: Missingness Examples

Sparses (r)	**Gaps** (n, λ)	**Stressed** (r, h, p)	**Rate**
0.001	(20, 0.1)	(0, 0.1, 0.99)	0.001
0.01	(200, 0.1)	(0.005, 0.5, 0.99)	0.01
0.05	(1000, 0.1)	(0.04, 0.2, 0.95)	0.05
0.1	(1000, 0.05)	(0.05, 0.5, 0.9)	0.1
0.3	(2000, 0.03)	(0.2, 1, 0.9)	0.3

Figure 2.2: Missingness parameters

2.2 Models

2.2.1 Deletion

The baseline of complete-case analysis: every date containing at least one missing value is dropped, and all downstream estimation runs on the surviving complete rows. Under MCAR the retained dates are a random subsample of the panel, so moment estimates remain consistent (Little and Rubin, 2019); the cost is statistical efficiency, and in the cross-section it is severe. With sparse missingness at rate r, a date survives with probability $(1-r)^N$: at $r = 0.05$ and $N = 50$ assets, fewer than 8% of the dates survive. Under the stressed pattern the damage changes nature rather than size: the deleted dates are exactly the turbulent ones, so the surviving subsample is calm by construction and every risk measure computed from it is biased downwards, no matter how much data is left.

2.2.2 Mean Replacement

The simplest imputation that preserves the shape of the panel consists in setting each missing return to its unconditional expected value. This is almost equivalent to setting the missing price to be equal to the last observed price, since daily returns are very close to zero. The mean of the panel is preserved, but every second moment is damped: under sparse missingness at rate r, a variance term survives whenever its entry is observed while a covariance term needs both entries at once, so estimated volatilities scale by $\sqrt{1-r}$ and pairwise correlations by $(1-r)$, shrinking the covariance matrix towards its diagonal. Under the stressed pattern the distortion is stronger still, because the entries being replaced are on average the largest ones. This predictable failure is precisely what the structural metrics of Section 2.3 are designed to expose.

2.2.3 Low Rank Matrix Completion

A few pervasive factors describe most of the covariance of equity returns, so the return panel is close to a low-rank matrix plus idiosyncratic noise. Matrix completion turns this observation into an imputation device: if the underlying matrix has rank $q \ll N$, the missing entries can be recovered from the observed ones – exactly so, under conditions on the missingness pattern and on how spread out the factors are (Candès and Recht, 2009). We adopt the iterative-SVD scheme of Mazumder et al. (2010): standardise each asset on its observed entries, fill the missing cells with zeros, then alternate between projecting the completed panel onto the set of rank-q matrices (truncated SVD) and restoring the observed entries, until the panel stops changing. The fixed point fills every missing cell with a q-factor reconstruction estimated on the fly, without ever specifying the factors. The nuclear-norm variant of the same algorithm, which soft-thresholds the singular values instead of truncating them, shrinks every imputed entry towards zero and thereby inherits a milder version of the covariance damping of mean replacement; we therefore keep the hard rank constraint, with q small, in line with the number of pervasive equity factors.

2.2.4 EM

Our likelihood-based model treats the dates as i.i.d. draws $r_t \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ and maximises the resulting observed-data likelihood with the EM algorithm of Dempster et al. (1977). Write $r_t = (r_{t,o}, r_{t,m})$ for the observed and missing parts of date t (the split changes from date to date). The E-step fills each missing block with its conditional expectation,

$$\mathbb{E}\left[r_{t,m} \mid r_{t,o}\right] = \boldsymbol{\mu}_m + \boldsymbol{\Sigma}_{mo}\, \boldsymbol{\Sigma}_{oo}^{-1} \left(r_{t,o} - \boldsymbol{\mu}_o\right),$$

while the M-step re-estimates $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}$ from the completed panel, adding back the conditional covariance $\boldsymbol{\Sigma}_{mm} - \boldsymbol{\Sigma}_{mo}\boldsymbol{\Sigma}_{oo}^{-1}\boldsymbol{\Sigma}_{om}$ of the cells just filled – the correction that separates EM from naively iterating a regression fill, which would understate the covariance. At convergence we obtain the maximum likelihood estimate $(\hat{\boldsymbol{\mu}}, \hat{\boldsymbol{\Sigma}})$ and, as a by-product, imputations that are exactly the L^2 projections discussed in Section 2.4. In the implementation, dates sharing the same missingness pattern are grouped together, so each pattern costs a single linear solve per iteration. Daily returns are of course heavier-tailed than the Gaussian assumption wants; the same iteration extends to a multivariate Student-t, whose E-step gains a per-date weight, decreasing in the Mahalanobis distance of the observed block, that discounts extreme dates in the update of $(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ (Little, 1988; Liu and Rubin, 1995).

2.2.5 KNN

k-nearest-neighbours imputation, first popularised on gene-expression data by Troyanskaya et al. (2001), is the donor-based, fully non-parametric member of our set. To fill date t, its distance to every other date is computed as the Euclidean distance over the assets both dates observe, rescaled by the number of co-observed assets so that sparser rows are not mechanically favoured; the missing return $r_{t,i}$ is then the inverse-distance-weighted average of $r_{s,i}$ over the k closest dates s that do observe asset i. Nothing about the scheme is temporal: donors are selected by cross-sectional similarity, wherever they sit in the sample. No distributional assumption is needed, but the method degrades exactly where donors become unreliable: under heavy missingness the distances rest on few co-observed assets, and under the stressed pattern the extreme dates that most need filling have no close, well-observed counterparts to borrow from.

2.2.6 BRITS

BRITS (Cao et al., 2018) is our recurrent neural representative. A window of the panel is processed by two LSTMs, one per time direction. At each step the hidden state is first decayed by a factor learned from the time elapsed since each asset was last observed, so that the memory of a stale reading fades, and two candidate fills are produced: a history-based estimate regressed from the hidden state, and a cross-sectional estimate regressed from the other assets at the same date. A learned gate blends the two, the blend replaces the missing entries, and the completed vector is fed to the next step, so the imputations themselves carry gradients through the network. Training is self-supervised: the network learns to reconstruct the entries that are observed, with a penalty tying the forward and backward passes together, and the missing cells are then read out of the trained network. No explicit model of the returns or of the missingness mechanism is assumed.

2.2.7 CSDI

CSDI (Tashiro et al., 2021) adapts denoising diffusion models (Ho et al., 2020) to imputation. In a diffusion model, data is progressively corrupted by Gaussian noise over many small steps until only noise remains, and a network is trained to undo one corruption step at a time; generation then amounts to running the learned chain backwards from pure noise. CSDI makes this chain conditional: the denoising network – a transformer applying attention along time and across assets (Vaswani et al., 2017) – always sees the clean observed entries while denoising the missing ones. Training is self-supervised, hiding a random share of the observed cells and learning to denoise them given the rest. At inference the chain starts from noise on the missing cells, with the observed ones held fixed throughout, and each pass produces one draw from the predictive distribution of the missing block given the observed panel; the final imputation is the median over several draws. CSDI is thus the one genuinely probabilistic model of our set – the dispersion of its draws is a built-in estimate of the imputation uncertainty, which no other model here provides.

2.3 Evaluation Methods

Every experiment produces, for a given missingness pattern and level, an imputed panel $\widehat{R} \in \mathbb{R}^{T\times N}$ to be compared against the ground truth R of Section 2.1. Let $\mathcal{M} = \{(t,k) : r_{t,k} \text{ was deleted}\}$ denote the set of imputed cells and $|\mathcal{M}|$ its cardinality. Since our stated goal is not the recovery of individual values but of the panel a downstream model would have been trained on, we score the imputations with two families of metrics. *Pointwise* metrics are computed on the imputed cells alone – the observed cells are returned untouched by every model, and including them would only dilute the error – and measure how close each filled value is to the deleted one. *Structural* metrics instead compare statistics computed from the two full panels, and measure whether the objects that downstream models actually consume (covariances, tail quantiles, marginal distributions, temporal dependence) survive the imputation. The two families need not agree: a method can be nearly optimal pointwise and still destroy the second-moment structure, as mean replacement shows. Deletion is the exception in what follows: it returns a panel with fewer rows, so the pointwise metrics are not defined for it and the structural ones are computed on the surviving subsample.

1. **MAE**: the mean absolute error on the imputed cells,

$$\mathrm{MAE} = \frac{1}{|\mathcal{M}|} \sum_{(t,k)\in\mathcal{M}} |\hat{r}_{t,k} - r_{t,k}|\,,$$

expressed in return units and robust to occasional large misses.

2. **RMSE**: the root mean squared error on the same cells,

$$\mathrm{RMSE} = \left(\frac{1}{|\mathcal{M}|} \sum_{(t,k)\in\mathcal{M}} (\hat{r}_{t,k} - r_{t,k})^2 \right)^{1/2}.$$

This is the sample counterpart of the L^2 distance of Section 2.4: the imputation minimising its expectation is the conditional expectation given the observed data, so RMSE rewards exactly the projections that the likelihood-based models construct, while penalising large errors quadratically.

3. **Covariance error**: the relative Frobenius error of the sample covariance matrix,

$$\mathrm{CovErr} = \frac{\|\widehat{\boldsymbol{\Sigma}} - \boldsymbol{\Sigma}\|_F}{\|\boldsymbol{\Sigma}\|_F},$$

where $\boldsymbol{\Sigma}$ and $\widehat{\boldsymbol{\Sigma}}$ are estimated from the true and the imputed panel respectively. This is our central structural metric: the covariance matrix is the object whose geometry (Section 2.4) we set out to preserve, and the input of essentially every allocation and risk model built downstream.

4. **Value-at-Risk error**: for each asset the empirical VaR at level $\alpha = 0.05$ is the (sign-flipped) empirical α-quantile of its return series, $\mathrm{VaR}_\alpha(k) = -\hat{q}_\alpha(R_k)$, and we report

$$\mathrm{VaRErr} = \frac{1}{N} \sum_{k=1}^{N} \left| \widehat{\mathrm{VaR}}_\alpha(k) - \mathrm{VaR}_\alpha(k) \right|.$$

This probes the left tail beyond the second moment, the region that risk management actually regulates – and precisely the region that the stressed pattern of Section 2.1.1 empties out.

5. **Wasserstein error**: the mean 1-Wasserstein distance between the per-asset marginal return distributions,

$$\text{WassErr} = \frac{1}{N}\sum_{k=1}^{N} W_1\left(\widehat{F}_k, F_k\right), \qquad W_1(F,G) = \int_{\mathbb{R}} |F(x) - G(x)|\, dx,$$

with F_k and $\widehat{F}_k$ the empirical distribution functions of asset k in the true and imputed panel. W_1 measures the minimal average displacement needed to transport one distribution onto the other; it compares the marginals in full, in return units, and detects shape distortions – e.g. imputed values overly concentrated near zero – that moment-based metrics summarise away.

6. **Absolute-return ACF error**: every metric above is invariant to a permutation of the dates; the last two are not. Daily returns are themselves close to serially uncorrelated, but their absolute values are persistently and positively autocorrelated – volatility clustering, the most robust temporal stylized fact of asset returns (Cont, 2001). Writing $\hat{\rho}_k(\cdot)$ for the sample autocorrelation at lag k, we compare the autocorrelation functions of the absolute returns up to a horizon K,

$$\text{AbsACFErr} = \frac{1}{NK}\sum_{j=1}^{N}\sum_{k=1}^{K}\left|\hat{\rho}_k\left(|\widehat{R}_j|\right) - \hat{\rho}_k(|R_j|)\right|,$$

with $K = 20$ throughout. An imputer that fills the panel with values that are individually plausible but temporally independent – as any model that treats the dates as exchangeable must – erases this dependence and is caught here and nowhere above.

7. **Leverage error**: negative returns raise subsequent volatility more than positive ones of the same size, the leverage effect (Black, 1976): the cross-correlation curve

$$L_k = \operatorname{corr}(r_t, |r_{t+k}|), \qquad k = 1, \ldots, K,$$

is negative at short lags. We report the mean absolute gap between the true and imputed curves, averaged over assets and lags as above. Unlike the previous metric, which is blind to sign, this one probes an *asymmetric* dependence between the level of a return and the volatility that follows it, and thereby whether the imputed panel still distinguishes falling from rising markets.

All seven metrics are computed by a single evaluation routine for every model, missingness pattern and missingness level; the resulting tables are collected and discussed in

Chapter 4.

2.4 Some Geometry

In this section we want to develop some geometrical intuition behind the evaluation methods criteria and the imputation models, as well as why a missing data problem requires different methods from an usual inference problem.

Take a multivariate time series of daily log returns:

$$R = \begin{pmatrix} r_1^\top \\ r_2^\top \\ \vdots \\ r_T^\top \end{pmatrix} = \begin{pmatrix} r_{11} & r_{12} & \cdots & r_{1N} \\ r_{21} & r_{22} & \cdots & r_{2N} \\ \vdots & \vdots & \ddots & \vdots \\ r_{T1} & r_{T2} & \cdots & r_{TN} \end{pmatrix} \in \mathbb{R}^{T\times N}$$

The (t)-th observation is:

$$r_t = (r_{t1}, \dots, r_{tN})^\top \in \mathbb{R}^N$$

The (i)-th variable asset realization is:

$$R_i = (r_{1i}, \dots, x_{Ti})^\top \in \mathbb{R}^T$$

Let us first consider each asset separately. We interpret each asset return series as a finite-sample realization of an underlying square-integrable random variable in a Hilbert space:

$$R_i \in L^2(\Omega, \mathcal{F}, \mathbb{P}) = \left\{X : \Omega \to \mathbb{R} : \mathbb{E}[X^2] < \infty\right\}$$

Square integrability, which is a reasonable assumption [1], enables us to define an useful inner product on L^2, namely:

$$\langle X, Y\rangle =: \mathbb{E}[XY]$$

This is remarkably similar to the definition of covariance, and is equal when the mean of X and Y is zero; at the daily level this is almost the case. We can use this approximation to see that the covariance and correlation of two assets are a function of their angle in L^2:

$$\langle X, Y\rangle \simeq \operatorname{Cov}(X, Y)$$

$$\rho(X, Y) = \frac{\operatorname{Cov}(X, Y)}{\operatorname{Var}(X)\operatorname{Var}(Y)} \simeq \frac{\langle X, Y\rangle}{\|X\|\|Y\|} = \cos(\theta)$$

The crucial part of this thesis is making sure that this linear geometry is preserved when imputing the missing data. The covariance matrix is of course the main object of

[1] Empirical work as Plerou et al. (1999) shows that the power law exponent of individual stocks is around 3, meaning variance exists but skew and kurtosis may not.

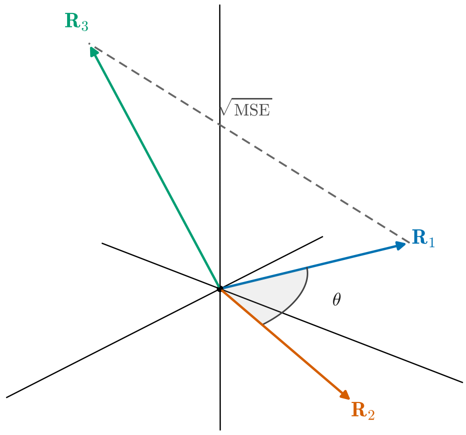

Figure 2.3: Returns as vectors in L^2

discussion, especially since it is used so widely across finance. It is immediate that MSE is the squared distance of two assets. The other metrics like VaR have no easy interpretation in this space.

Figure 2.3 is an attempt to represent this abstract space. Now let us turn to the realized space $\mathbb{R}^T$: the definitions above still apply, bar the due discretizations, but now we can represent the missing data as a projection of the full data. Write:

$$R_i = P_{\mathcal{O}_i} R_i + P_{\mathcal{M}_i} R_i = R_{O_i} + R_{M_i}$$

Where $P_{\mathcal{O}_i}$ is the operator that deletes the missing data and $P_{\mathcal{M}_i}$ is the one that keeps it and deletes the observed one. Their ranges are clearly orthogonal under the usual inner product. The goal of our models is to approximate R_{M_i} with:

$$\hat{R}_{M_i} = h_\theta(R_{O_i}, R_{O_{k \neq i}})$$

If we knew the values of R_{M_i} we could run an approximation in the least squares sense. Given h_θ, find:

$$\min_\theta \|R_{M_i} - \hat{R}_{M_i}\|$$

This is essentially equivalent to trying to approximate the conditional expectation, or in geometric terms projecting R_{M_i} onto the space of functions $\sigma(R_{O_i}$ and $R_{O_{k \neq i}})$, a subspace of L^2.

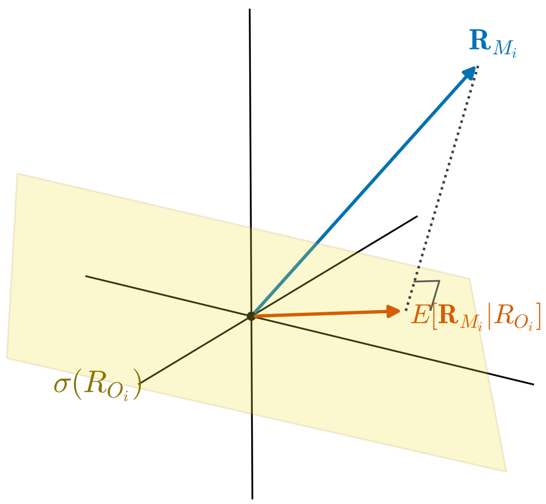

Figure 2.4: Conditional expectation as a projection

Chapter 3

Imputation Methods

Chapter 4

Results and Discussion

Tables 4.1, 4.2 and 4.3 report the seven evaluation metrics of Section 2.3 for every model, one table per injection pattern of Section 2.1.1, with the levels ordered by their realised share of missing cells. Within each level, the best value of every metric is set in bold. A dash marks a value that is not defined: the pointwise errors of Deletion (the returned panel changes shape), its remaining metrics at the heaviest levels (too few complete dates survive), and KNN at the most extreme stressed level, where the dates to be filled lack well-observed donors altogether.

Three regularities stand out. First, the hierarchy is stable across patterns and levels: the conditional methods dominate the two baselines everywhere, EM is rarely beaten – and never by much – while the neural models match it at a far higher computational cost rather than surpassing it (BRITS even stumbles at two stressed levels). Second, pointwise and structural quality diverge exactly as anticipated in Section 2.3: under the MCAR patterns mean replacement stays within roughly fifty per cent of the best MAE while its covariance error is close to an order of magnitude worse than EM's. Third, MNAR is the genuinely hostile regime: at the heaviest stressed level no method brings the relative covariance error below 0.6 – the turbulent dates are simply no longer in the data, and no choice of model recovers them.

Table 4.1: Evaluation scores under sparse missingness (MCAR); levels are the deletion rate r.

Level	Model	MAE	RMSE	CovErr	VaRErr	WassErr	AbsACFErr	LevErr
0.001	Deletion	–	–	0.0119	0.0001	0.0001	0.0101	0.0114
	Zero	0.0118	0.0194	0.0026	**0.0000**	**0.0000**	0.0009	0.0008
	KNN	0.0099	0.0168	0.0018	**0.0000**	**0.0000**	0.0006	0.0005
	LowRank	0.0084	0.0152	0.0016	**0.0000**	**0.0000**	**0.0005**	**0.0004**
	EM	**0.0080**	**0.0147**	**0.0015**	**0.0000**	**0.0000**	**0.0005**	**0.0004**
	BRITS	0.0091	0.0158	0.0016	**0.0000**	**0.0000**	**0.0005**	**0.0004**
	CSDI	0.0081	0.0151	0.0016	**0.0000**	**0.0000**	**0.0005**	**0.0004**
0.01	Deletion	–	–	0.0532	0.0005	0.0004	0.0247	0.0334
	Zero	0.0107	0.0161	0.0205	0.0001	0.0001	0.0035	0.0027
	KNN	0.0081	0.0123	0.0069	0.0001	**0.0000**	0.0019	0.0017
	LowRank	0.0077	0.0117	0.0054	**0.0000**	**0.0000**	0.0016	0.0015
	EM	0.0071	0.0109	0.0051	**0.0000**	**0.0000**	**0.0015**	**0.0014**
	BRITS	0.0076	0.0116	0.0064	0.0001	**0.0000**	0.0017	0.0016
	CSDI	**0.0069**	**0.0107**	**0.0050**	**0.0000**	**0.0000**	**0.0015**	**0.0014**
0.05	Deletion	–	–	0.1951	0.0019	0.0012	0.1334	0.0515
	Zero	0.0111	0.0167	0.1044	0.0005	0.0006	0.0132	0.0073
	KNN	0.0084	0.0127	0.0283	0.0003	0.0002	0.0047	0.0045
	LowRank	0.0078	0.0118	0.0169	**0.0002**	**0.0001**	0.0045	0.0041
	EM	**0.0072**	0.0113	0.0151	**0.0002**	0.0002	0.0044	**0.0039**
	BRITS	0.0073	0.0113	**0.0145**	**0.0002**	0.0002	**0.0043**	0.0040
	CSDI	**0.0072**	**0.0112**	0.0149	**0.0002**	0.0002	0.0051	0.0041
0.1	Deletion	–	–	0.4492	0.0091	0.0051	–	–
	Zero	0.0109	0.0163	0.1872	0.0011	0.0011	0.0276	0.0111
	KNN	0.0083	0.0125	0.0487	0.0007	0.0004	0.0090	0.0067
	LowRank	0.0077	0.0116	0.0282	**0.0004**	**0.0003**	0.0072	0.0058
	EM	**0.0071**	**0.0110**	**0.0237**	0.0005	**0.0003**	**0.0068**	**0.0056**
	BRITS	0.0072	0.0111	0.0238	0.0005	**0.0003**	**0.0068**	0.0057
	CSDI	0.0072	0.0111	0.0261	**0.0004**	**0.0003**	0.0077	0.0057
0.3	Deletion	–	–	–	–	–	–	–
	Zero	0.0109	0.0161	0.4749	0.0037	0.0033	0.0638	0.0201
	KNN	0.0087	0.0129	0.1467	0.0024	0.0014	0.0173	0.0118
	LowRank	0.0080	0.0119	0.0749	**0.0014**	**0.0007**	0.0170	0.0105
	EM	**0.0074**	**0.0112**	0.0565	0.0017	0.0009	0.0150	**0.0099**
	BRITS	0.0076	0.0114	**0.0536**	0.0015	0.0008	**0.0149**	0.0101
	CSDI	**0.0074**	0.0113	0.0641	**0.0014**	0.0008	0.0203	0.0112

Table 4.2: Evaluation scores under contiguous-gap missingness (MCAR); levels are the parameters (n, λ) of Section 2.1.1.

Level	Model	MAE	RMSE	CovErr	VaRErr	WassErr	AbsACFErr	LevErr
(20, 10)	Deletion	–	–	0.0103	0.0001	0.0001	0.0033	0.0033
	Zero	0.0121	0.0184	0.0027	**0.0000**	**0.0000**	0.0003	0.0002
	KNN	0.0096	0.0144	0.0016	**0.0000**	**0.0000**	0.0002	**0.0001**
	LowRank	0.0087	0.0123	0.0010	**0.0000**	**0.0000**	0.0002	**0.0001**
	EM	0.0079	0.0117	0.0011	**0.0000**	**0.0000**	**0.0001**	**0.0001**
	BRITS	**0.0078**	**0.0112**	0.0010	**0.0000**	**0.0000**	**0.0001**	**0.0001**
	CSDI	0.0079	0.0115	**0.0009**	**0.0000**	**0.0000**	**0.0001**	**0.0001**
(200, 10)	Deletion	–	–	0.1492	0.0010	0.0004	0.0243	0.0158
	Zero	0.0107	0.0156	0.0222	**0.0001**	0.0001	0.0037	0.0015
	KNN	0.0082	0.0121	0.0065	**0.0001**	**0.0000**	0.0016	0.0012
	LowRank	0.0078	0.0118	0.0061	**0.0001**	**0.0000**	0.0015	0.0012
	EM	**0.0072**	**0.0109**	**0.0046**	**0.0001**	**0.0000**	**0.0014**	**0.0011**
	BRITS	0.0073	0.0111	0.0049	**0.0001**	**0.0000**	0.0015	**0.0011**
	CSDI	**0.0072**	**0.0109**	0.0049	**0.0001**	**0.0000**	0.0015	**0.0011**
(1000, 10)	Deletion	–	–	0.1825	0.0017	0.0010	0.1135	0.0493
	Zero	0.0104	0.0152	0.0785	0.0004	0.0005	0.0155	0.0048
	KNN	0.0080	0.0118	0.0224	0.0003	0.0002	0.0059	0.0032
	LowRank	0.0074	0.0108	0.0137	**0.0002**	**0.0001**	**0.0048**	0.0029
	EM	**0.0069**	**0.0103**	0.0116	**0.0002**	**0.0001**	0.0049	0.0028
	BRITS	0.0070	0.0104	0.0116	**0.0002**	**0.0001**	0.0049	0.0028
	CSDI	**0.0069**	**0.0103**	**0.0114**	**0.0002**	**0.0001**	0.0051	**0.0027**
(1000, 20)	Deletion	–	–	0.8524	0.0087	0.0042	0.1872	0.1681
	Zero	0.0110	0.0163	0.1716	0.0010	0.0010	0.0354	0.0068
	KNN	0.0085	0.0127	0.0487	0.0007	0.0004	0.0117	0.0049
	LowRank	0.0078	0.0119	0.0310	**0.0004**	**0.0003**	0.0088	0.0046
	EM	**0.0073**	**0.0114**	**0.0238**	0.0005	**0.0003**	0.0083	**0.0044**
	BRITS	0.0074	**0.0114**	0.0241	0.0005	**0.0003**	**0.0082**	0.0045
	CSDI	**0.0073**	0.0115	0.0328	**0.0004**	**0.0003**	0.0095	**0.0044**
(2000, 30)	Deletion	–	–	1.4799	0.0104	0.0100	–	–
	Zero	0.0108	0.0161	0.4324	0.0028	0.0027	0.0908	0.0143
	KNN	0.0085	0.0128	0.1378	0.0018	0.0011	0.0277	0.0090
	LowRank	0.0080	0.0120	0.0820	**0.0010**	**0.0006**	0.0218	0.0083
	EM	**0.0073**	**0.0113**	0.0589	0.0012	0.0007	0.0212	**0.0079**
	BRITS	0.0074	0.0114	**0.0555**	0.0011	**0.0006**	**0.0187**	0.0082
	CSDI	0.0074	0.0115	0.0896	**0.0010**	0.0007	0.0265	0.0091

Table 4.3: Evaluation scores under stressed missingness (MNAR); levels are the parameters (r, h, p) of Section 2.1.1.

Level	Model	MAE	RMSE	CovErr	VaRErr	WassErr	AbsACFErr	LevErr
(0, 0.1, 0.99)	Deletion	–	–	0.2813	0.0009	0.0004	0.0582	0.0200
	Zero	0.0552	0.0640	0.0676	0.0001	0.0001	0.0115	0.0070
	KNN	0.0202	0.0285	0.0229	**0.0000**	**0.0000**	0.0034	0.0029
	LowRank	0.0158	0.0220	0.0116	**0.0000**	**0.0000**	0.0025	0.0020
	EM	**0.0133**	**0.0187**	**0.0094**	**0.0000**	**0.0000**	**0.0021**	**0.0017**
	BRITS	0.0301	0.0385	0.0253	**0.0000**	**0.0000**	0.0046	0.0036
	CSDI	0.0163	0.0226	0.0175	**0.0000**	**0.0000**	0.0027	0.0022
(0.005, 0.5, 0.99)	Deletion	–	–	0.2960	0.0010	0.0005	0.0687	0.0268
	Zero	0.0321	0.0463	0.2238	0.0005	0.0003	0.0483	0.0172
	KNN	0.0161	0.0261	0.0991	0.0001	**0.0001**	0.0168	0.0095
	LowRank	0.0129	0.0200	**0.0279**	**0.0000**	**0.0001**	0.0078	0.0067
	EM	**0.0118**	**0.0189**	0.0319	**0.0000**	**0.0001**	**0.0077**	**0.0065**
	BRITS	0.0122	0.0197	0.0401	**0.0000**	**0.0001**	0.0082	0.0069
	CSDI	0.0136	0.0233	0.0751	**0.0000**	**0.0001**	0.0132	0.0087
(0.04, 0.2, 0.95)	Deletion	–	–	0.5325	0.0043	0.0018	0.1079	0.0491
	Zero	0.0143	0.0219	0.2254	0.0010	0.0007	0.0375	0.0143
	KNN	0.0092	0.0141	0.0562	0.0003	0.0002	0.0092	0.0067
	LowRank	0.0083	0.0125	0.0253	**0.0002**	**0.0001**	0.0068	0.0052
	EM	**0.0077**	**0.0117**	**0.0217**	**0.0002**	**0.0001**	**0.0062**	**0.0048**
	BRITS	0.0078	0.0118	0.0231	**0.0002**	**0.0001**	0.0064	0.0049
	CSDI	0.0078	0.0122	0.0374	**0.0002**	**0.0001**	0.0078	0.0061
(0.05, 0.5, 0.9)	Deletion	–	–	0.6318	0.0056	0.0023	0.1230	0.0630
	Zero	0.0174	0.0249	0.5125	0.0030	0.0017	0.0843	0.0244
	KNN	0.0103	0.0161	0.1602	0.0010	0.0004	0.0258	0.0124
	LowRank	0.0091	0.0139	**0.0483**	**0.0004**	**0.0002**	0.0110	0.0091
	EM	**0.0084**	**0.0130**	0.0554	0.0008	0.0003	**0.0107**	**0.0087**
	BRITS	0.0106	0.0162	0.2083	0.0015	0.0006	0.0256	0.0126
	CSDI	0.0086	0.0137	0.0542	0.0008	0.0003	0.0165	0.0109
(0.2, 1, 0.9)	Deletion	–	–	–	–	–	–	–
	Zero	0.0149	0.0219	0.7677	0.0070	0.0042	0.1201	0.0333
	KNN	–	–	–	–	–	–	–
	LowRank	0.0137	0.0212	0.6525	**0.0062**	0.0029	0.1111	0.0316
	EM	**0.0134**	**0.0210**	0.6634	0.0065	0.0031	0.1112	0.0315
	BRITS	0.0136	0.0214	**0.6480**	0.0063	**0.0025**	0.1069	0.0299
	CSDI	**0.0134**	**0.0210**	0.6559	0.0064	0.0028	**0.1065**	**0.0298**

4.1 Computational costs

Chapter 5

Conclusion

5.1 Summary

5.2 Limitations

5.3 Future Work

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